

Hypergeometrical Distribution:

In binomial experiment, the trials are independent and the probability of success remains constant from trial and trial. So, the binomial distribution can be applied to a sampling with replacement. But, if ~~an~~ in an experiment, the prob. of success does not remain in the same from one trial to another, the Binomial distribution cannot be applied. In this case, an appropriate distribution is hypergeometrical distribution.

An experiment of a sampling without replacement from a finite population is known as hypergeometrical experiment.

Definition:

A random variable X is said to be followed a hypergeometrical distribution with three parameters, N , M and n if it assumes only non-negative values and its prob is;

$$p(X=x) = \frac{\binom{M}{x} \binom{N-M}{n-x}}{\binom{N}{n}} \quad \begin{matrix} , x=0, 1, 2, \dots, \\ \min(n, M) \\ , 0 \quad , \text{otherwise} \end{matrix}$$

where, N is the population size (finite), M is the total number of items with a certain characteristic in the population and n is the sample size, where $M \leq N$ and $n \leq N$.

Derivation of Hypergeometric Distribution:

Let N be the total number of objects and M of them (N) have a certain characteristic, called M successes and $N-M$ objects don't have the characteristic called $N-M$ failures.

Let ' n ' be the sample size that is drawn from N objects without replacement. Let x be the number of successes that is drawn from M successes in $\binom{M}{x}$ ways and $(n-x)$ failures can be drawn from the remaining $N-M$ failures in $\binom{N-M}{n-x}$ ways. Then the probability of selecting x successes from M successes and $(n-x)$ failures from $N-M$ failures is given by;

$$p(x=x) = \frac{\binom{M}{x} \binom{N-M}{n-x}}{\binom{N}{n}} \quad / \quad x=0,1,2,\dots,\min(M,n)$$

This is the prob. mass function (pmf) of Hypergeometric distribution with parameters N, M and n . That will verify such way.

$$\sum_{x=0}^n p(x=x)$$

$$= \sum_{x=0}^n \frac{\binom{M}{x} \binom{N-M}{n-x}}{\binom{N}{n}}$$

$$= \frac{1}{\binom{N}{n}} \left[\sum_{x=0}^n \binom{M}{x} \binom{N-M}{n-x} \right]$$

$$= \frac{1}{\binom{N}{n}} \times \binom{N}{n}$$

$$= 1$$

$$\therefore \sum_{x=0}^n p(x=x) = 1$$

Moment of Hypergeometric Distribution:

Let X be a hypergeometric random variable with its pmf

$$P(X=x) = \frac{\binom{M}{x} \binom{N-M}{n-x}}{\binom{N}{n}}, \quad x=0,1,2,\dots,\min(n,M)$$

The first raw moment about origin is given as;

$$H_1' = E(X) = \sum x p(x=x)$$

$$= \sum_{x=0}^n x \frac{\binom{M}{x} \binom{N-M}{n-x}}{\binom{N}{n}}$$

$$= \frac{1}{\binom{N}{n}} \sum_{x=0}^n x \binom{M}{x} \binom{N-M}{n-x}$$

$$= \frac{1}{\binom{N}{n}} \sum_{x=0}^n x \frac{M!}{x!(M-x)!} \binom{N-M}{n-x}$$

$$= \frac{1}{\binom{N}{n}} \sum_{x=1}^n x \frac{M(M-1)!}{x(x-1)!(M-x)!} \binom{N-M}{n-x}$$

$$= \frac{M}{\binom{N}{n}} \left[\sum_{x=1}^n \frac{(M-1)!}{(x-1)!(M-x)!} \right] \binom{N-M}{n-x}$$

$$\therefore \sum_{x=0}^n \binom{a}{x} \binom{b}{n-x} = \binom{a+b}{n}$$

$$= \frac{M}{\binom{N}{n}} \sum_{x=1}^n \binom{M-1}{x-1} \binom{N-M}{n-x}$$

$$= \frac{M}{\binom{N}{n}} \times \binom{N-1}{n-1}$$

$$= \frac{M}{N C_n} \times \frac{N-1}{n-1}$$

$$E(x) = n \left(\frac{M}{N} \right) = \frac{nM}{N}$$

Again,

$$E(x^2) = \sum_{x=0}^n x^2 p(x=x)$$

$$= \sum_{x=0}^n [x(x-1) + x] \frac{\binom{M}{x} \binom{N-M}{n-x}}{\binom{N}{n}}$$

$$= \sum_{x=0}^n x(x-1) \frac{\binom{M}{x} \binom{N-M}{n-x}}{\binom{N}{n}} + \sum_{x=0}^n x \frac{\binom{M}{x} \binom{N-M}{n-x}}{\binom{N}{n}}$$

$$= \frac{1}{\binom{N}{n}} \sum_{x=0}^n x(x-1) \frac{M!}{x!(M-x)!} \times \binom{N-M}{n-x} + \frac{nM}{N}$$

$$= \frac{1}{\binom{N}{n}} \sum_{x=2}^n x(x-1) \frac{M(M-1)(M-2)!}{x(x-1)(x-2)!(M-x)!} \times \binom{N-M}{n-x} + \frac{nM}{N}$$

$$= \frac{M(M-1)}{\binom{N}{n}} \sum_{x=2}^n \frac{(M-2)!}{(x-2)!(M-x)!} \times \binom{N-M}{n-x} + \frac{nM}{N}$$

$$E(x^2) = \frac{M(M-1)}{\binom{N}{n}} \left[\sum_{x=2}^n \binom{M-2}{x-2} \binom{N-M}{n-x} \right] + \frac{nM}{N}$$

$$E(x^2) = \frac{M(M-1)}{\binom{N}{n}} \times \binom{N-2}{n-2} + \frac{nM}{N}$$

$$E(x^2) = \frac{M(M-1)}{N!} \times \frac{(N-2)!}{(n-2)!(N-n+2)!} + \frac{nM}{N}$$

$$= \frac{M(M-1)n!(N-n)!}{N!} \times \frac{(N-2)!}{(n-2)!(N-n+2)!} + \frac{nM}{N}$$

$$= \frac{M(M-1)n!(N-2)!}{N!(n-2)!} + \frac{nM}{N}$$

$$E(x^2) = \frac{M(M-1)n(n-1)}{N(N-1)} + \frac{nM}{N}$$

now,

$$V(x) = E(x^2) - [E(x)]^2$$

$$= \frac{M(M-1)n(n-1)}{N(N-1)} + \frac{nM}{N} - \frac{n^2 M^2}{N^2}$$

$$= \frac{nM}{N} \left[\frac{(M-1)(n-1)}{(N-1)} + 1 - \frac{nM}{N} \right]$$

$$\boxed{V(x) = \frac{nM(N-M)}{N^2(N-1)}}$$

Show that the hypergeometric distribution tends to Binomial distribution as $N \rightarrow \infty$ and $\frac{M}{N} = p$.

Proof:

Let x be a r.v that follows hypergeometric distribution (HGD) and its pmf is given by as;

$$P(X=x) = \frac{\binom{M}{x} \binom{N-M}{n-x}}{\binom{N}{n}}$$

$$= \frac{M!}{x!(M-x)!} \times \frac{(N-M)!}{(n-x)!(N-M-n+x)!} \times \frac{n!(N-n)!}{N!}$$

$$= \frac{M(M-1)(M-2)\dots(M-x+1)(M-x)!}{x!(M-x)!} \times \frac{(N-M)(N-M-1)\dots(N-M-n+x+1)(N-M-n+x)!}{(n-x)!(N-M-n+x)!} \times \frac{n!(N-n)!}{N(N-1)\dots(N-n+1)(N-n)!}$$

$$= \frac{n!}{x!(n-x)!} \left(\frac{M(M-1)(M-2)\dots(M-x+1)}{N(N-1)(N-2)\dots(N-x+1)} \times \frac{(N-M)(N-M-1)\dots(N-M-n+x+1)}{N(N-1)(N-2)\dots(N-n+1)} \times \frac{n!}{N(N-1)\dots(N-n+1)} \right)$$

$$= \frac{n!}{x!(n-x)!} \left(\frac{M(M-1)(M-2)\dots(M-x+1)}{N(N-1)(N-2)\dots(N-x+1)} \times \frac{(N-M)(N-M-1)\dots(N-M-n+x+1)}{N(N-1)(N-2)\dots(N-n+1)} \times \frac{n!}{N(N-1)\dots(N-n+1)} \right)$$

$$\lim_{n \rightarrow \infty} P(X=x) = \frac{n!}{x!(n-x)!} \left(\frac{M}{N} \left(\frac{M}{N} - \frac{1}{N} \right) \dots \left(\frac{M}{N} - \frac{x-1}{N} \right) \times \frac{(N-M)}{N} \left(\frac{N-M}{N} - \frac{1}{N} \right) \dots \left(\frac{N-M}{N} - \frac{n-x-1}{N} \right) \times \frac{n!}{N(N-1)\dots(N-n+1)} \right)$$

$$= \frac{n!}{x!(n-x)!} \lim_{N \rightarrow \infty} \left(\frac{M}{N} \left(\frac{M}{N} - \frac{1}{N} \right) \dots \left(\frac{M}{N} - \frac{x-1}{N} \right) \times \frac{(N-M)}{N} \left(\frac{N-M}{N} - \frac{1}{N} \right) \dots \left(\frac{N-M}{N} - \frac{n-x-1}{N} \right) \times \frac{n!}{N(N-1)\dots(N-n+1)} \right)$$

$$= \frac{n!}{x!(n-x)!} \lim_{N \rightarrow \infty} p \cdot p \dots p \times (1-p)(1-p) \dots (1-p)$$

$$= \frac{n!}{x!(n-x)!} p^x q^{n-x}$$

$$\lim_{N \rightarrow \infty} p(x=r) = {}^n C_r p^r q^{n-r}$$

Negative Hypergeometric Distribution:

Let x be a random variable that denotes the number of trials required to get a fixed number of objects with specified characteristic is called negative hypergeometric random variable and its distribution is called negative hypergeometric distribution.

Definition:

A random variable x is said to be followed a negative hypergeometric distribution with three parameters N, M and r and its pmf is;

$$p(x=r) = \frac{{}^{x-1} C_{r-1} {}^{N-x} C_{M-r}}{{}^N C_M}, \quad x=r, r+1, r+2, \dots, N$$

Derivation of NHGD:

Let a population consists of N objects out of which M objects have a certain characteristic is called success (say defective) and $(N-M)$ objects do not have the characteristic called failure (say non-defective).

Let x be a random variable that denotes the number of draws needed to draw r th success in sampling without replacement. If the draws are made until r th success is obtained then the r th success is obtained in the last x th draw with probability $\frac{M-(r-1)}{N-(x-1)}$ and $(r-1)$ successes are obtained in the previous draws with probability $\frac{{}^M C_{r-1} {}^{N-M} C_{x-r}}{{}^N C_{x-1}}$ where $x \geq r$